

PLC SYLLABUS

5-Day Intensive (Codesys / TIA Portal)

DAY	CORE TOPICS	INSTRUCTIONS / BLOCKS COVERED	SOFTWARE FOCUS	GOAL / PROJECT
DAY 1	PLC architecture, scan cycle, memory areas, I/O addressing, contact logic	NO/NC contacts, OUT, SET/RESET, seal-in (latching) logic, edge detection (R_TRIG/F_TRIG)	Codesys ladder editor, TIA Portal LAD	Motor Start/Stop with Seal-In Circuit — includes E-stop interlock and indicator lamp logic
DAY 2	Timers, counters, sequencing basics	TON, TOF, TP, RTO/CTU, CTD, CTUD, step-sequencer pattern	Codesys, TIA Portal timer blocks	Timer & Counter Program — traffic-light sequence or conveyor cycle counter with reset logic
DAY 3	Analog I/O, scaling, data handling	Raw-to-engineering unit scaling (SCALE_X, NORM_X), arithmetic (ADD/SUB/MUL/DIV), comparators (GT/LT/EQ), INT/REAL/BOOL/WORD data types	Codesys analog config, TIA Portal tag tables	Temperature Monitoring System — analog sensor input, threshold alarms, high/low limit indication
DAY 4	Structured programming, control loops	Function Blocks (FB) vs Functions (FC), instance data blocks, shift registers, PID block tuning (P/I/D gains)	Codesys POU's, TIA Portal FB/FC/DB structure	Batch Mixing Station — multi-stage sequencing (fill → mix → drain) with PID-controlled agitator speed
DAY 5	Industrial networking, visualization	Modbus TCP/RTU addressing, HMI tag linking, alarm/event logging, basic SCADA screens	Codesys Modbus config, TIA Portal HMI/WinCC	Capstone: Sorting & Packaging Line — sensor-triggered sorting logic, HMI control panel, Modbus data exchange between PLC and HMI

NOTES FOR DEPTH (OPTIONAL ADD-ONS PER DAY)

- Day 1: Add motor forward/reverse interlock as a stretch goal.
- Day 2: Add a shift-register-based bottle sequencing demo for lead-in to Day 4.
- Day 3: Cover signal noise/filtering basics (moving average) if time allows.
- Day 4: Compare FB (stateful) vs FC (stateless) explicitly — common interview question.
- Day 5: If Modbus setup is slow, fall back to OPC UA tag browsing demo instead.